INSTITUTE OF MATHEMATICAL SCIENCES, FACULTY OF SCIENCE, UNIVERSITI MALAYA
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TUTORIAL 10

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1. The natural length of a spring is 1 meter. An object is attached to it and the length of the spring increases to 1.02 meter when the object is in equilibrium. Then the object is initially displaced 1 cm below the equilibrium position and given an upward velocity of 14 cm s^{-1}. Use the acceleration due to gravity, $g = 9.8$ cm s^{-2}, find the equation of motion.

2. A spring with spring constant 20 N/m is attached to a 2 kg mass with the damping constant is 4 times the instantaneous velocity. If the mass has initial position 1 meter below the equilibrium and released with no initial velocity, find the equation of motion.

3. A 1 kg mass is attached to a spring whose constant is 16 N/m and the entire system is then submerged in a liquid that imparts a damping force numerically equal to 10 times the instantaneous velocity. The mass is initially released from rest from a point 1 meter below the equilibrium position.
   (a) Find the equation of motion.
   (b) Determine whether the mass passes through the equilibrium position.

4. A spring with spring constant (stiffness) 16 N/m is attached to a 1 kg mass. Given that the damping constant is 10 times the instantaneous velocity. The mass is initially released from a point 1 meter below the equilibrium position with an upward velocity of 12 m/s.
   (a) Find the equation of motion.
   (b) Determine the time at which the mass passes through the equilibrium position.